Contextro: Token-Efficient Repository Discovery for MCP-Based Coding Workflows

Paired evaluation on an 8,496-file production monorepo

Distillation Labs

ABSTRACT

Repository discovery is a major source of token overhead in MCP-based coding workflows. This report evaluates Contextro, a proprietary local code-intelligence server for repository search, symbol navigation, structural analysis, and persistent memory, using a 60-task paired study on an 8,496-file production TypeScript monorepo. Relative to a lexical baseline based on grep plus bounded file reads, the Contextro pipeline reduced total token consumption from 328,272 to 14,407, reduced median latency from 448.35 ms to 5.21 ms, and eliminated file reads. On the directly comparable 39-task subset, token consumption fell to 6,638, a 98.0% reduction; across 100 deterministic bootstrap subsets, reduction remained between 97.24% and 98.48%. Public reruns on the open `src/` tree reproduced low-token and high-retrieval-quality behavior. These results describe discovery efficiency rather than end-to-end coding correctness.

Keywords: Model Context Protocol, repository discovery, code intelligence, semantic retrieval, structural search, technical evaluation

1. Introduction

Repository discovery remains a costly phase in MCP-based coding workflows. Before a client can act on the correct symbol, file, or interface, it typically executes multiple broad searches, opens several files, and accumulates context that is only partially relevant to the task. In the paired study reported here, a representative lexical workflow consumed 5,524 tokens and took 1.11 s, whereas the corresponding indexed `find_symbol` call returned in 5.66 ms with 65 tokens.

Contextro addresses this inefficiency by replacing file-by-file exploration with structured repository access. It exposes repository search, symbol navigation, relationship tracing, impact analysis, structural search, history lookup, and persistent memory through a local MCP server. The operating premise is straightforward: when discovery tasks can be answered directly, the client does not need to spend context budget reconstructing repository structure from raw files.

This report makes three bounded claims. First, a local code-intelligence layer can materially reduce the token cost of repository discovery. Second, the resulting interface can deliver low latency on a production-scale repository for common lookup tasks. Third, the effect remains visible under stricter analysis that separates directly comparable discovery tasks from MCP-native capabilities with no lexical analogue.

2. Related Work

This work sits within a broader literature on contextual retrieval, hybrid search, and prompt-budget management. Prior work from Anthropic, NVIDIA, Liu et al., and OpenAI argues that retrieval systems perform best when they combine multiple signals, rank relevant evidence near the top of the context window, and avoid transmitting unnecessary text. The present report applies that general logic to repository discovery, where the cost of locating the right code often dominates the cost of reasoning over it.

3. System Design

Contextro operates as a local MCP service positioned between the coding client and the repository. Rather than requiring exploratory file reads, it returns structured answers to discovery questions, including identifiers, locations, relationships, and compact supporting excerpts. The objective is to shift discovery from file traversal toward targeted retrieval.

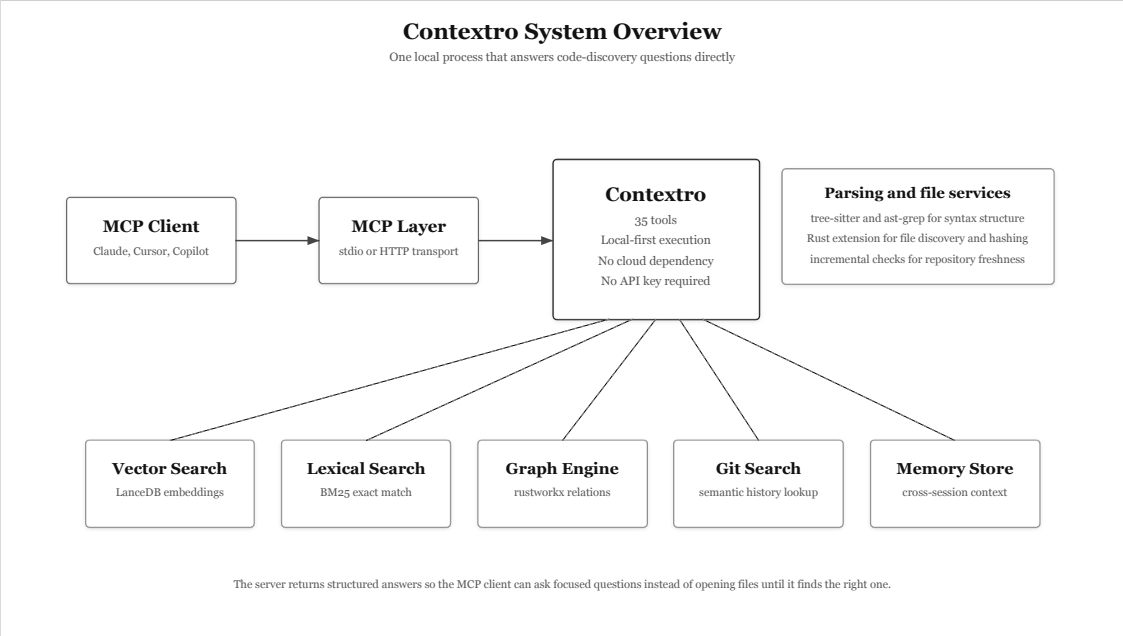

Figure 1. System overview. Contextro exposes a unified local interface for repository search, structural analysis, history lookup, and memory.

3.1. System Components

Table 1. Capability map.

Capability	Representative tools	Returned information
Search by meaning	`search`	Exact snippet, file, and line
Find a symbol	`find_symbol`	Definition location and caller count
Trace relationships	`find_callers`, `find_callees`, `explain`	Callers, callees, and local context
Assess blast radius	`impact`	Transitive impact set
Search structure	`code(pattern_search, ...)`	AST-level matches
Analyze reachability	`dead_code`, `circular_dependencies`	Structural analysis results
Search history	`commit_search`	Semantically relevant commits
Persist context	`remember`, `recall`	Durable memory across sessions

3.2. Retrieval and Response Design

The retrieval layer combines semantic, lexical, and structural signals. Semantic retrieval supports concept-level matching, lexical retrieval supports exact identifiers and strings, and graph-based traversal supports questions about callers, callees, and impact. Returned material is filtered and abbreviated so that the interface favors direct answers over large context transfers.

The implementation is local and single-process. It integrates vector indexing, lexical retrieval, structural analysis, syntax-aware search, history lookup, and memory in one service boundary. That integration matters operationally because it eliminates the need to orchestrate several external systems during interactive discovery.

4. Methodology

The evaluation uses a paired two-condition design on a single production repository. Each task is executed once with a lexical discovery workflow and once with the Contextro interface. The full task inventory contains 60 tasks across 16 categories. Among these, 39 tasks have direct lexical analogues and therefore support strict paired comparison; the remaining 21 tasks represent MCP-native capabilities with no meaningful file-centric baseline.

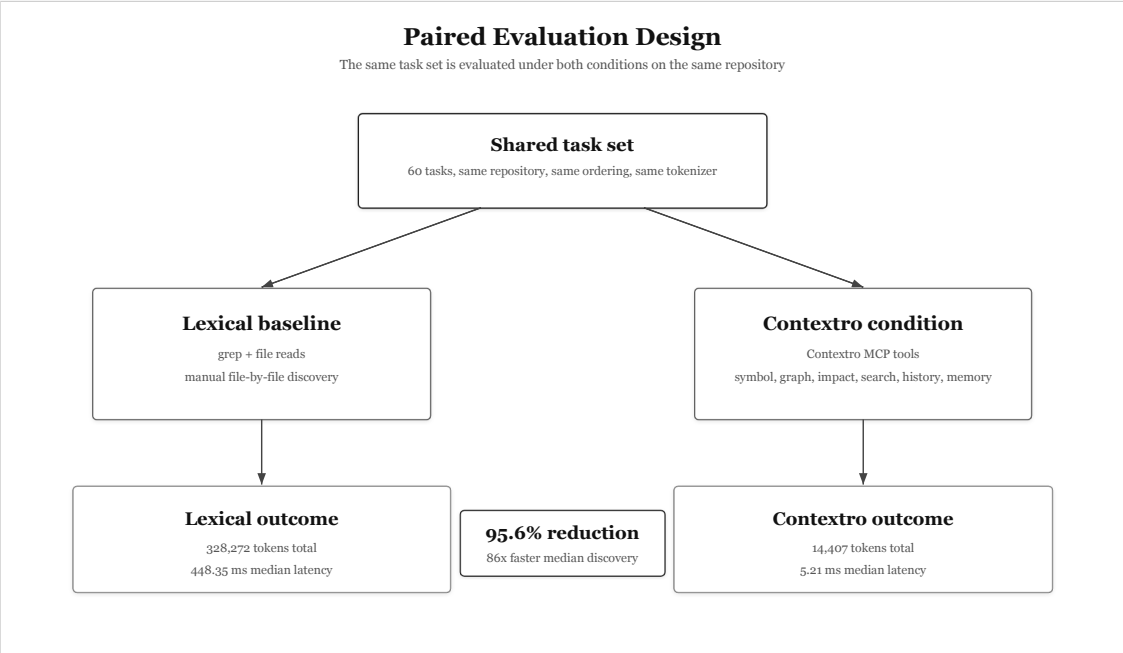

Figure 2. Study design. The same task set is evaluated under both conditions on the same repository, preserving a paired comparison.

4.1. Study Design

The lexical baseline uses repository-wide grep plus bounded follow-up file inspection. For each directly comparable task, the baseline includes grep output and the contents of up to five matching files in the token estimate. The Contextro condition executes one MCP tool call against a pre-indexed repository state. Tasks without a meaningful lexical analogue are retained in the full inventory as MCP-native capabilities, but they are excluded from the directly comparable subset and from the bootstrap robustness analysis.

This design isolates discovery cost rather than downstream coding quality. It compares a file-centric lexical workflow with a structured retrieval interface under a fixed task inventory, shared repository, and common token accounting rule.

4.2. Repository Under Test

Table 2. Repository under test.

Property	Value
Repository	Private production TypeScript monorepo
Indexed files	8,496 files in the private production monorepo
Languages	TypeScript and JavaScript
Task inventory	60 tasks across 16 categories
Directly comparable subset	39 tasks across 8 categories

Property	Value
MCP-native tasks	21 tasks without meaningful lexical analogues
Indexing observation	21.1 s full indexing time in the fresh rerun
Verification host	Apple M4, 16 GB RAM, macOS 26.4.1, Python 3.12.13

4.3. Metrics and Reporting

Table 3. Metrics and reporting.

Measurement	Definition	Artifact
Token estimate	Common tokenizer applied to baseline outputs and Contextro responses	Aggregate study summary and rerun summary
Latency	Wall-clock milliseconds measured around each task invocation	Aggregate study summary and rerun summary
Tool calls and files read	Count of baseline grep/read steps versus one MCP tool call	Aggregate study summary and rerun summary
Comparable subset	All tasks with a direct lexical analogue	Comparable-task catalog and subset robustness analysis
Robustness analysis	100 bootstrap task subsets of size 39 with replacement	Subset robustness analysis
Open benchmark verification	Public reruns on the open `src/` tree for token and retrieval checks	Public benchmark artifacts

The study is deterministic rather than stochastic in the usual machine-learning sense, so reporting emphasizes medians, ranges, and subset robustness rather than significance testing. The bootstrap analysis is used as a sensitivity check on task composition, not as a population-level inference over repositories.

5. Results

5.1. Headline Outcomes

Across the full 60-task inventory, total token consumption fell from 328,272 in the lexical baseline to 14,407 in the Contextro condition. Mean tokens per task fell from 5,471.2 to 240.1, median latency fell from 448.35 ms to 5.21 ms, and mean files read fell from 2.5 per task to 0. **The lexical baseline models bounded file-centric discovery rather than a full autonomous agent policy. The reported gains therefore characterize discovery efficiency, not end-to-end coding correctness.**

The stricter comparison is the 39-task subset with direct lexical analogues. On that subset, token use fell to 6,638, a 98.0% reduction, while median latency fell from 498.89 ms to 4.89 ms and mean tool calls fell from 4.9 to 1.0. The gap between the 60-task and 39-task summaries reflects the presence of 21 MCP-native tasks that expose analysis surfaces absent from the lexical baseline.

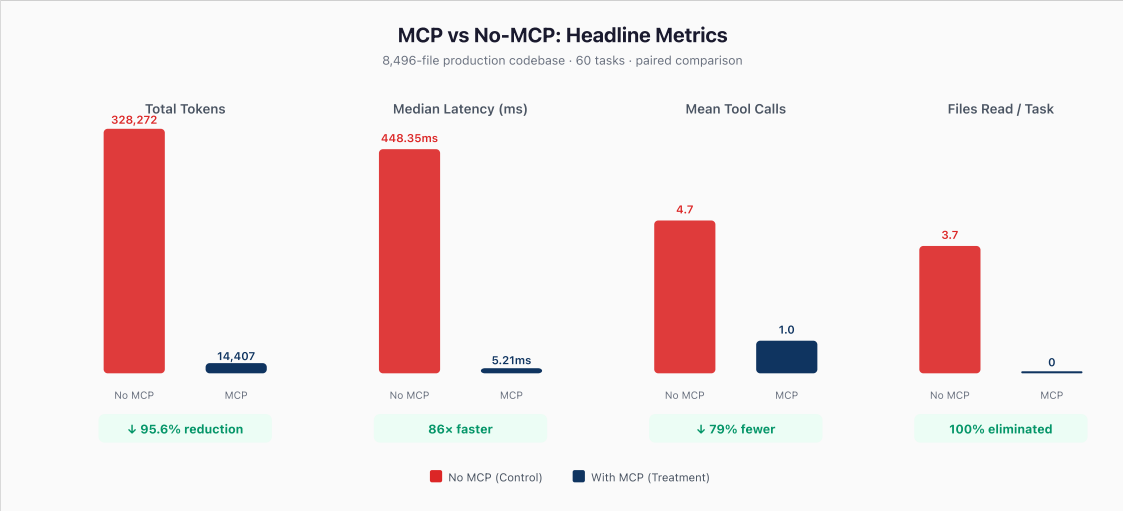

Figure 3. Headline outcomes from the paired study. The figure summarizes the full task inventory; the stricter directly comparable subset is reported separately in Table 4.

Table 4. Headline outcomes.

Analysis set	Tasks	MCP-native tasks	Lexical tokens	Contextro tokens	Token result	Median latency
Full task inventory	60	21	328,272	14,407	95.6% reduction	448.35 ms → 5.21 ms
Directly comparable subset	39	0	328,272	6,638	98.0% reduction	498.89 ms → 4.89 ms
100 bootstrap subsets	100 × 39	0	—	—	median 97.87% 97.24%-98.48%	lexical 458.28-525.76 ms Contextro 2.15-7.00 ms

No bootstrap subset fell below 97.24% token reduction. The effect therefore remains stable across many task mixtures drawn from the directly comparable subset.

5.2. Results by Category

Table 5 reports the eight categories with direct lexical analogues. Semantic search and symbol discovery show the largest reductions at 98.8%. Caller tracing, impact analysis, code understanding, git history search, and the comparable portion of the `code` tool remain above 96%.

Tail latency is concentrated in a small number of heavier operations. The `focus` category is bimodal because one request returns in 4.81 ms while another deep file-context request takes 7.01 s. The comparable `code` category also contains a repository-wide `pattern_search` outlier at 11.22 s.

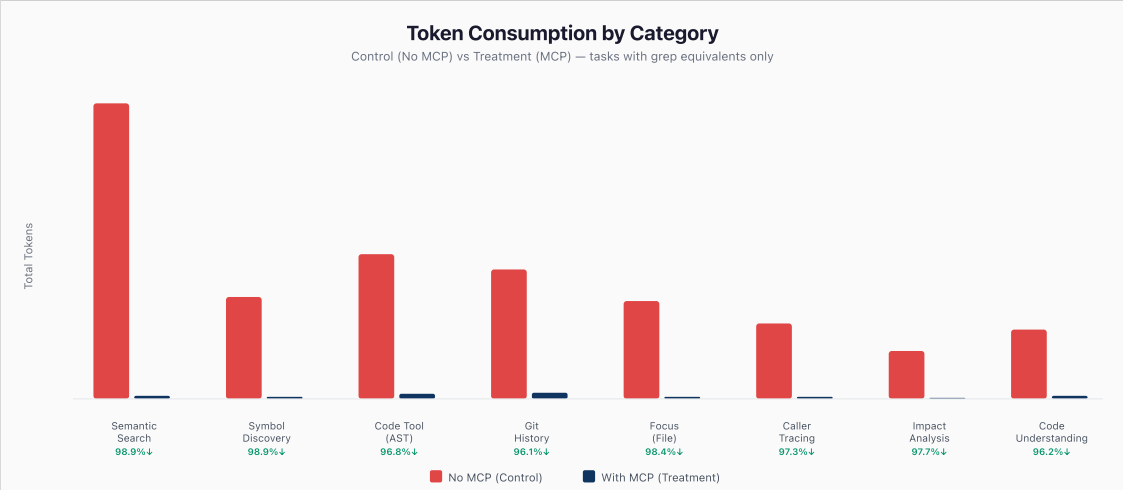

Figure 4. Results by task category. The visualization focuses on discovery categories with direct lexical analogues.

Table 5. Results by category.

Category	Comparable tasks	Lexical tokens	Contextro tokens	Reduction	Contextro latency
Semantic search	8	96,366	1,199	98.8%	131.53 ms median 95.19-138.31 ms
Symbol discovery	8	34,481	413	98.8%	1.81 ms median 1.37-9.13 ms
Caller/callee tracing	6	25,915	730	97.2%	1.46 ms median 0.92-4.89 ms
Impact analysis	4	16,225	377	97.7%	0.93 ms median 0.90-3.68 ms
Code understanding	6	23,526	944	96.0%	6.68 ms median 6.22-8.26 ms
Git history search	2	46,106	1,117	97.6%	6.51 ms median 5.21-7.80 ms
Code tool (comparable slice)	3	51,459	1,318	97.4%	1.92 ms median 0.97 ms-11.22 s
Focus / file context	2	34,194	540	98.4%	3.51 s median 4.81 ms-7.01 s

5.3. Tool Latency

Most lookup and graph operations remain in the low-millisecond range. The long tail is composed of a small number of deliberately heavier analyses rather than broad slowness. In the fresh rerun, `focus_01` took 7.01 s and repository-wide `code(pattern_search, ...)` took 11.22 s; both are preserved explicitly in Appendix C rather than absorbed into category means.

Table 6. Representative tool latencies.

Tool	Fresh rerun latency	Role
`health`	0.88 ms	Server status
`status`	1.47 ms	Server status

Tool	Fresh rerun latency	Role
`find_callers`	1.97 ms median	Graph lookup
`find_symbol`	1.81 ms median	Symbol lookup
`impact`	0.93 ms median	Blast radius analysis
`explain`	6.68 ms median	Local understanding
`commit_search`	6.51 ms median	History search
`overview`	93.24 ms	Project summary
`search`	131.53 ms median	Hybrid retrieval
`architecture`	436.75 ms	Deep structural summary
`dead_code`	1,307.18 ms	Reachability analysis
`code(pattern_search, ...)`	11.22 s	Repository-wide AST search outlier

5.4. Public Verification

Public benchmark reruns on the open `src/` tree reproduce the low-token and high-retrieval-quality behavior reported in the paired study. These public checks do not substitute for the private monorepo evaluation, but they do provide an observable external reference for token output and retrieval quality on a redistributable codebase.

Table 7. Optimization and verification artifacts.

Artifact	Measurement	Result	Source
Historical workflow benchmark	16-call workflow token budget	2,552 → 1,043 total output tokens (-59.1%)	Historical tuning log
Historical indexing benchmark	Full index and no-change incremental reindex	279 s → 6.8 s full index; 22 ms no-change incremental	Historical tuning log
Public token rerun	16-call public workflow benchmark	1,286 total output tokens; 16 calls; 7.5% token reduction	Public benchmark artifact
Public retrieval rerun	20 query retrieval benchmark on `src/`	Hybrid MRR 1.0, Recall@1 1.0, Recall@5 1.0, 169.2 avg tokens	Public benchmark artifact

6. Discussion

The most important interpretive point is that the result becomes stronger, not weaker, under the stricter directly comparable subset. The full 60-task inventory shows a 95.6% token reduction, while the 39-task subset shows a 98.0% reduction because it excludes MCP-native capabilities that have no lexical baseline. This separation clarifies the distinction between replacing file-centric discovery and adding entirely new analysis surfaces.

The bootstrap robustness analysis strengthens that interpretation. Across 100 deterministic task subsets, reduction never fell below 97.24% and the median remained 97.87%. These values do not establish universal generalization across repositories, but they do indicate that the result is not driven by a small number of unusually favorable tasks.

Latency should be interpreted similarly. Common lookup operations are effectively instantaneous, while a few repository-wide analyses remain expensive. The practical result is not uniform sub-10 ms performance; it is that routine discovery operations remain fast while deeper structural queries continue to return structured answers at substantially lower token cost than exploratory file reads.

7. Limitations

The findings in this report are bounded. They describe repository discovery efficiency under a paired task protocol, not general-purpose coding quality under all workflows or repositories.

Table 8. Study limitations.

Area	Issue	Status
Paired-study codebase availability	The original 8,496-file monorepo is private and cannot be redistributed with the paper	Mitigated by shipping the harness, sanitized task inventory, rerun summary, and public benchmark reruns
Control baseline approximation	The lexical baseline models bounded file-centric discovery with grep plus up to five file reads, not a full autonomous agent policy	Interpret as a discovery-cost baseline rather than a universal baseline for agent quality; end-to-end correctness benchmark is future work
Tail latency	`focus_01` and repository-wide `pattern_search` dominate the latency tail at 7.01 s and 11.22 s	Reported explicitly in Appendix C and excluded from any misleading latency simplification
Public proxy repository	The paired study corpus is a private monorepo; the public benchmark reruns use only the 78-file `src/` tree	A public proxy repository (≥1,000 files) for full rerun reproducibility is planned as future work
End-to-end correctness	The study measures discovery cost (tokens, latency, file reads), not downstream coding correctness	An end-to-end edit-task correctness benchmark is future work
Next.js App Router reachability	`dead_code` flags live routes as unreachable	Known limitation
Duplicate symbol disambiguation	Multiple definitions with the same name can conflate callees	Under investigation
Complexity metrics	`maintainability_index` reports 0 for TypeScript	Not yet implemented for TS

The study evaluates one production TypeScript monorepo and focuses on discovery efficiency, latency, and file-read behavior. It does not report an end-to-end correctness benchmark for code changes, nor does it compare against stronger alternatives such as LSP-backed discovery or syntax-aware lexical tooling. The claims in this report should therefore be read as claims about discovery performance and interface quality.

8. Availability and Reproducibility

The public artifact package distinguishes between materials that can be rerun on public code and materials that remain conditional because the study corpus is private. It includes sanitized task catalogs, aggregate study summaries, subset robustness outputs, and public benchmark reruns. **Contextro is a proprietary product of Distillation Labs and is not distributed as open-source software.**

8.1. Artifact Package

Artifact class	Public contents
Task definitions	Sanitized machine-readable catalogs for the 60-task full inventory and the 39-task directly comparable subset
Aggregate study outputs	Archived and fresh paired-study summaries, plus the 100-subset robustness analysis
Public reruns	Token-efficiency and retrieval-quality reruns on the public `src/` tree

Artifact class	Public contents
Evaluation harness	Scripts required to reproduce public benchmarks and to rerun the paired study where an equivalent private repository is available

8.2. Rerunnable Commands (Public Benchmarks)

```
pip install -e .[dev]

python scripts/benchmark_token_efficiency.py
python scripts/benchmark_retrieval_quality.py --path src --query-limit 20 --output
docs/publication/open-retrieval-quality.json

ruff check .
pytest -v -m "not slow"
```

8.3. Conditional Paired-Study Rerun

The full paired study requires access to the original private monorepo or to an adapted equivalent repository. The published catalogs preserve task counts, categories, and tool coverage without disclosing private repository identifiers. Where an equivalent repository is available locally, the rerun protocol is:

```
python scripts/experiment_platform.py --output-dir docs/publication --export-task-catalogs-only
python scripts/experiment_platform.py --codebase /path/to/private-monorepo --output-dir
/secure/local-output
python scripts/task_subset_robustness.py --results /secure/local-output/results.json --subsets 100
--output docs/publication/paired-study-subset-robustness.json
```

8.4. Verification Environment

The reruns cited in this report were executed on Apple silicon hardware using macOS 26.4.1, an Apple M4 CPU, 16 GB RAM, and Python 3.12.13. The public benchmark harness supports Python 3.10-3.12.

9. Conclusion

Contextro demonstrates that a local code-intelligence layer can reduce the token and latency cost of repository discovery in MCP-based coding workflows. In the paired evaluation reported here, the full 60-task inventory reduced total tokens by 95.6%, while the stricter 39-task directly comparable subset reduced them by 98.0% and remained above 97.24% across 100 bootstrap task subsets. The primary conclusion is narrow but durable: structured repository access can materially outperform bounded file-centric discovery for repository search and navigation.

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Appendix A. Experiment Configuration

Appendix A summarizes the paired-study configuration at a high level. It is included to document study scope and reporting parameters without exposing implementation details beyond what is necessary for interpretation.

Parameter	Value
Total task inventory	60 tasks across 16 categories
Directly comparable subset	39 tasks with lexical analogues
MCP-native tasks	21 tasks with no meaningful lexical baseline
Token accounting	Common tokenizer applied to baseline outputs and Contextro responses
Lexical baseline	Repository-wide grep plus bounded follow-up file inspection
Robustness procedure	100 bootstrap task subsets of size 39 with replacement
Verification host	macOS 26.4.1, Apple M4, 16 GB RAM, Python 3.12.13
Public supporting artifacts	Sanitized task catalogs, aggregate summaries, robustness outputs, and public benchmark reruns

Appendix B. MCP Tool API Contracts

This appendix summarizes the externally visible MCP tool contracts used by the evaluation. Parameter lists and response fields are described at the interface level. All tools return an MCP `ToolResult` whose `structured_content` field carries the JSON payload summarized below; errors are returned as `{"error": "…message…"}`.

B.1. Discovery and Navigation

`search(...)` returns ranked retrieval results with file locations, symbol labels, and compact code excerpts.

`find_symbol(...)` returns exact or fuzzy symbol matches, including definition metadata and local relationship counts.

`find_callers(...)` and `find_callees(...)` return direct relationship sets for the requested symbol.

`explain(...)` returns symbol metadata together with explanatory prose and supporting snippets.

`impact(...)` returns a transitive impact set summarizing affected symbols, files, and directories.

B.2. Project Intelligence

`overview()`, `architecture()`, `focus(...)`, and `restore()` return project- or file-level summaries intended to orient the client within the repository.

`analyze(...)`, `dead_code()`, `circular_dependencies()`, `test_coverage_map()`, and `audit()` return higher-level analysis summaries rather than direct retrieval results.

B.3. Git and History

`commit_search(...)` returns ranked commit matches for a query, while `commit_history(...)` returns chronological commit lists. `repo_status()` summarizes the multi-repository registry state.

B.4. Code Intelligence

`code(operation, ...)` dispatches syntax-aware operations such as symbol lookup, pattern search, and codebase overview generation. Output structure varies by operation but consistently returns result or match collections.

B.5. Memory and Session

`remember(...)`, `recall(...)`, and `forget(...)` manage persistent memory entries. `knowledge(...)`, `session_snapshot()`, and `compact(...)` manage broader session and knowledge state.

B.6. Server Utilities

`status()`, `health()`, and `index(...)` expose operational server state. `retrieve(...)` fetches sandboxed large responses by reference, and `introspect(...)` returns runtime tool schemas and settings.

Table 9. MCP tool API contracts summary.

Tool	Category	Primary inputs	Primary outputs
`search`	Discovery	query, limit, mode, language, symbol_type	total, results[symbol_name, score, filepath, line_start, code]
`find_symbol`	Discovery	name, exact	n, f, l, t, callers, callees (single) / total, symbols (multi)
`find_callers`	Discovery	symbol_name	callers[] (single) / definitions[{definition, callers}] (multi)
`find_callees`	Discovery	symbol_name	callees[]
`explain`	Discovery	symbol_name, verbosity	merged symbol fields, analysis, related_code
`impact`	Discovery	symbol_name, max_depth	total_impacted, impacted_symbols, impacted_files, impacted_dirs
`overview`	Intelligence	(none)	total_files, total_symbols, total_relationships, languages, directories
`architecture`	Intelligence	(none)	layers, entry_points, hub_symbols, top_classes
`focus`	Intelligence	path, include_code	role, symbols, imports, nearby_tests, blast_radius
`restore`	Intelligence	(none)	project, entry_points, layers, risk_summary
`analyze`	Intelligence	path	complexity, code_smells, quality
`dead_code`	Intelligence	(none)	summary, unused_files, unused_symbols
`circular_dependencies`	Intelligence	(none)	summary, cycles
`test_coverage_map`	Intelligence	(none)	summary, uncovered_files, uncovered_symbols
`audit`	Intelligence	(none)	summary, recommendations
`commit_search`	Git	query, limit, branch, author	total, results[hash, author, date, message, top_files]
`commit_history`	Git	path, limit, since	total, branch, commits[hash, author, date, message]
`repo_status`	Git	(none)	total_repos, total_files, repos, watcher
`code`	AST	operation, query/pattern/...	results or matches (shape varies by operation)
`remember`	Memory	content, memory_type, tags, ttl	id, status
`recall`	Memory	query, limit, memory_type, tags	total, memories[]
`forget`	Memory	memory_id, tags, memory_type	deleted_count
`session_snapshot`	Session	(none)	codebase, branch, head, vector_chunks, symbols, events
`compact`	Session	content	archived, archive_ref, chars, hint
`status`	Server	(none)	codebase_path, indexed, graph{n,r,f}
`health`	Server	(none)	status, uptime_seconds, indexed, engines
`index`	Server	path, paths	status, message, path

Tool	Category	Primary inputs	Primary outputs
`retrieve`	Server	ref_id, query	ref_id, content
`introspect`	Server	query, doc_path	tools, settings (full runtime schemas)

Appendix C. Paired-Study Results by Tool Family

Table 10 reports aggregate token and latency measurements grouped by tool family. Rows marked *No* in the lexical-equivalent column represent MCP-native capabilities and are excluded from the directly comparable subset and bootstrap analyses discussed in Section 5.

Table 10. Paired-study results by tool family.

Tool family	Category	Lexical equivalent?	Lexical tokens	Contextro tokens	Contextro latency (ms)	Reduction
find_symbol ×8	Symbol discovery	Yes	34,481	413	1.37–9.13	98.8%
find_callers/find_callees ×6	Caller tracing	Yes	25,915	730	0.92–4.89	97.2%
search ×8	Semantic search	Yes	96,366	1,199	95.19–138.31	98.8%
explain ×6	Code understanding	Yes	23,526	944	6.22–8.26	96.0%
impact ×4	Impact analysis	Yes	16,225	377	0.90–3.68	97.7%
focus ×2	File context	Yes	34,194	540	4.81–7,009.60	98.4%
commit_search ×2	Git history	Yes	46,106	1,117	5.21–7.80	97.6%
code ×5 (3 comparable / 2 native)	AST operations	Mixed	51,459	1,627	0.97–11,222.49	96.8%
commit_history ×1	Git history	No	—	919	93.18	N/A
overview	Project structure	No	—	85	93.24	N/A
architecture	Project structure	No	—	893	436.75	N/A
analyze	Code quality	No	—	1,128	29.21	N/A
dead_code	Static analysis	No	—	933	1,307.18	N/A
circular_dependencies	Static analysis	No	—	545	0.77	N/A
test_coverage_map	Static analysis	No	—	555	11.16	N/A
audit	Full report	No	—	994	72.52	N/A
remember/recall/forget	Memory	No	—	161	4.04–13.35	N/A
session_snapshot/restore/compact	Session	No	—	984	0.86–53.54	N/A
status/health	Server ops	No	—	107	0.88–1.47	N/A
introspect	Self-documentation	No	—	46	0.31	N/A
repo_status	Multi-repo	No	—	102	1.40	N/A

Tool family	Category	Lexical equivalent?	Lexical tokens	Contextro tokens	Contextro latency (ms)	Reduction
knowledge (show)	Knowledge base	No	—	8	0.47	N/A